



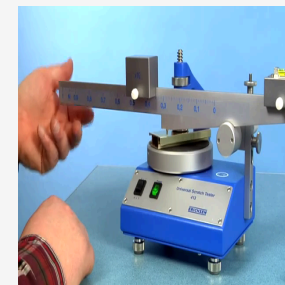
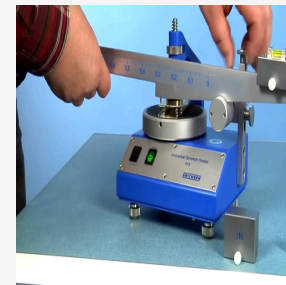
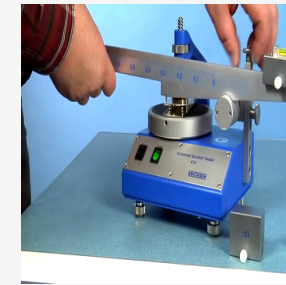
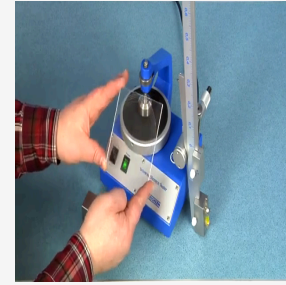
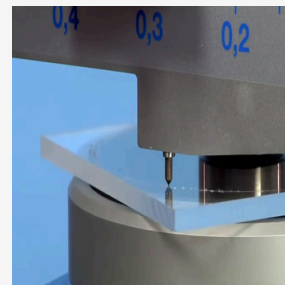
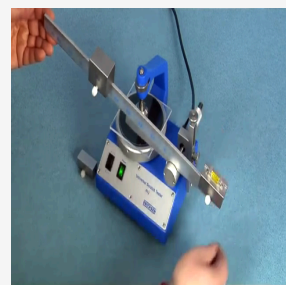
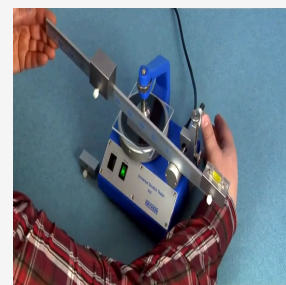
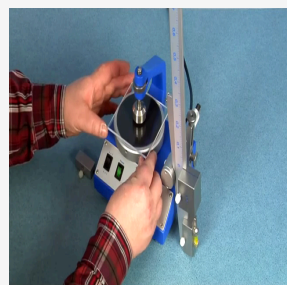
Original Video



Question (Temporal Grounding) a man is working with a blue metal scale



Ground Truth [start, end]



Ground Truth [51.2, 67.3]

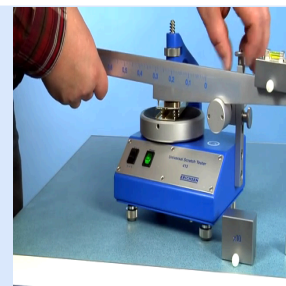
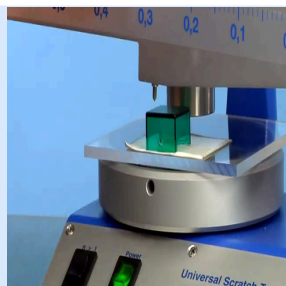
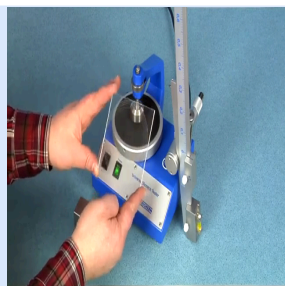
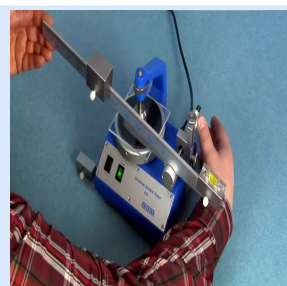
0s

Video Duration

70s



Sparse View (FPS=0.5)



Ground Truth [51.2, 67.3]

0s

$T_s \approx 35$ frames

70s

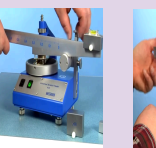
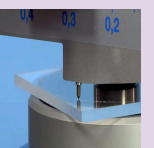
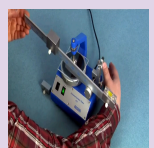


Question

a man is working with a blue metal scale



Dense View (FPS=2.0)



Ground Truth [51.2, 67.3]

0s

$T_d \approx 140$ frames

70s



Sparse Response

Thinking: The person is working with a blue metal scale. ... So the estimated time range is approximately from 49.2s to 68.1s.

Final Answer: [49.2,68.1]

Format Reward

R_{fmt}^s

Grounding Reward

R_{grd}^s

Reference Reward

$R_{\text{ref}} = w^d \cdot \text{tIoU}(\hat{\tau}^s, \hat{\tau}^d)$ where $w^d = \mathbb{I}[R_{\text{grd}}^d > R_{\text{grd}}^s]$



Policy Model



Dense Response

Thinking: The man is interacting with a blue metal scale. He first adjusts the device and prepares it for measurement. ... So the event can be localized precisely around 50.5s to 66.7s.

Final Answer: [50.5,66.7]

Grounding Reward

R_{grd}^d

Format Reward

R_{fmt}^d